

Testing Domain-Specific Email and Phone Detection for GDPR Compliance in Cloud Food Data Lakes

Zero False Positives, 99.3% Recall — Validated Against Microsoft Presidio

DEPLOYED IN **SOS FOOD** · HORIZON EUROPE GRANT NO. 101134894
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Outline

1. **Motivation** — why generic PII tools fail on food data
2. **Approach** — a three-layer pipeline and four metamorphic relations
3. **Evaluation** — precision/recall, a Presidio comparison, mutation testing
4. **Conclusion** — production impact, limitations, future work

Part 1

Motivation

GDPR Compliance Starts at Upload, Not After

The platforms

- FOODITY — GDPR-compliant, open-access European food data lake
- **SOS Food** — our project, extends it to closed-access settings (Horizon Europe, Jan 2024–Dec 2027, Grant No. 101134894)
- Providers upload CSV files up to **1 GB**
- Async pipeline: antivirus scan → **PII detection** → CSV parsing → bulk indexing

The stakes

Once a file is indexed, its content becomes discoverable — an email or phone number slipped into a "food safety measurements" file creates real legal liability.

If we block by mistake, providers abandon a 1 GB upload. If we miss real PII, it becomes searchable.

Barcodes Look Like Phones to Generic Tools

Why food data fools them

Food data is dense with 8–16 digit patterns that syntactically match international phone formats:

- GPS coordinates
- EAN-13 barcodes
- Timestamps, nutritional decimals
- Batch and classification codes

Presidio, Macie, and Google DLP go straight from **candidate extraction to validation** — no extension point to filter domain patterns first.

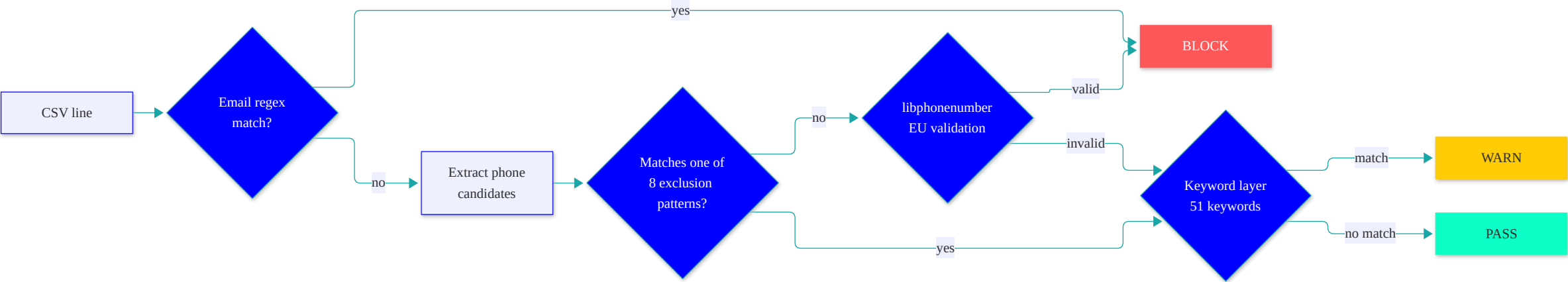
Our four design requirements

#	Requirement
R1	Zero false positives
R2	Reliable EU email/phone detection
R3	Graceful degradation via warnings
R4	Streaming — no full-file load

Part 2

Approach

Filter Before You Validate — Not After



$D(\ell) = \text{block if } E(\ell) \vee \exists c \in C(\ell): \neg X(c) \wedge V(c); \text{ warn if } K(\ell); \text{ else pass}$

ℓ one CSV line (the input)	$E(\ell)$ the line contains an e-mail address	$C(\ell)$ phone-like candidates extracted from the line	$X(c)$ candidate matches an exclusion pattern (date, GPS, barcode...) → it is food data	$V(c) \cdot K(\ell)$ V: valid European number · K: line contains a PII keyword
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Eight Patterns Encode 18 Months in Production

Pattern	Matches
Date-like	ISO/EU dates (2023-01-15, 16/06/2021)
Date-time-like	Timestamps (16/06/2021 16:53)
Coordinate tuple	GPS pairs (6.036 -4.978 54.0)
Hyphen-ID	Lab references (17024320-1817)
Slash-ID	Classification codes (000/000/000/17)
Slash-path	Hierarchical IDs (000/045/012)
Decimal number	Nutritional values (13.6857142857)
Leading zeros	EAN barcodes (0000000015707)

Metamorphic Testing: No Exact Oracle Needed

The problem — we cannot list the expected verdict for every possible CSV line (the *oracle problem*).

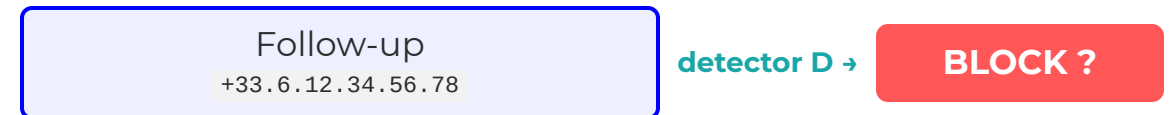
The idea — start from a *seed* whose verdict we know, apply a transformation that must **not** change it, and check that the verdict stays the same.

Two-sided seeds — 26 *positive* seeds must BLOCK (they protect recall); 52 *negative* seeds must PASS (they protect precision).

1 • Known verdict



2 • Transform (MR2: reformat) → verdict must not change



The same **78 seeds** (26 positive + 52 negative) are used twice:

- ① **Regression suite** — 4 relations generate **207 tests** (+ 22 unit tests, 1 corpus assertion)
- ② **Evaluation corpus** — seeds expanded over exclusion families, formats and case, + **28 real production samples = 500 samples** (200 pos., 300 neg.)

Context Must Not Change the Verdict (MR1, MR3)

MR1 · Positive context

PII hidden among ordinary food-data fields must still be caught.

$$D(p)=\text{block} \Rightarrow D(t_1\|p\|t_2)=\text{block}$$

Example (illustrative) — 3 food-data fields added around the PII:

seed	a.martin@mail.eu → BLOCK
follow-up	2023-01-15;13.6857142857; 00000000015707;a.martin@mail.eu → must still BLOCK

Generated by prepending/appending $k = 3$ fields drawn from the negative seeds to each positive seed.

MR3 · Negative context

Harmless food-data values combined must not accidentally look like a phone number.

$$D(f)=\text{pass} \Rightarrow D(c_1\|f\|c_2)=\text{pass}$$

Example (illustrative) — two values that each PASS:

value A	6.036 -4.978 54.0	→ PASS
value B	17024320-1817	→ PASS
A + B	6.036 -4.978 54.0;17024320-1817	→ must still PASS

Generated by combining every pair of the 12 production-derived rows: $C(12,2) = 66$ tests.

Format and Case Must Not Change the Verdict

MR2 · Phone format invariance

Writing the same number differently must not change the verdict.

$D(p)=\text{block} \Rightarrow D(\sigma(p))=\text{block}$

6 variants per phone seed (illustrative) — all must BLOCK:

+33 6 12 34 56 78	spaces
+33.6.12.34.56.78	dots
+33-6-12-34-56-78	hyphens
0033 6 12 34 56 78	00 prefix
+33 (6) 12 34 56 78	parentheses
+33612345678	compact

MR4 · Email case invariance

Upper/lower case must not change the verdict (RFC 5321).

$D(e)=\text{block} \Rightarrow D(\kappa(e))=\text{block}$

4 variants per email seed (illustrative) — all must BLOCK:

a.martin@mail.eu	lower
A.MARTIN@MAIL.EU	upper
A.Martin@Mail.Eu	mixed
a.martin@MAIL.EU	domain upper

Part 3

Evaluation

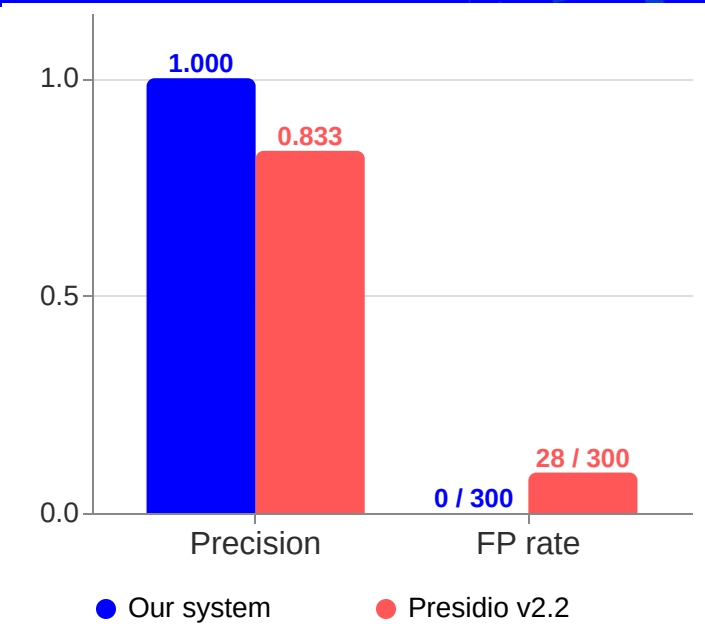
Perfect Precision, 99.3% Recall on 500 Samples

	Flagged	Passed
In-scope PII (n=141)	140	1
Out-of-scope PII (n=59)	0	59
No PII (n=300)	0	300

500 samples in total = 472 synthetic + **28 production-derived** (real uploads). That is 300 negatives (food data) + 200 positives (11 PII categories).



Zero False Positives vs. Presidio's 28



Same protocol, same data

- Same 500-sample corpus for both tools
- Presidio v2.2.362, default Phone/Email recognizers — no custom recognizer, no deny-list
- Same in-scope recall (0.993): both miss the same VOIP-classified number

What Presidio flagged

28 food values labelled `PHONE_NUMBER` (score 0.4):

Family	Example
GPS tuples	38.7223 -9.1393
Customs codes	0402 10 19
Hyphenated IDs	12345678-901

Same 300 food-data rows, 28 wrong flags vs 0 → McNemar exact test, $p \approx 7.5 \cdot 10^{-9}$. The errors cluster in 3 families, so no threshold tuning removes them.

Mutation Testing: 78% of Faults Killed

78%

Mutation score — full suite (57/73 killed)

60%

MR-derived tests alone (207 tests)

Of the 16 surviving mutants:

- **9** never covered (keyword layer, outside the block/pass oracle)
- **7** are covered equivalents (unreachable guards, shadowed patterns)

The mutation run itself **found the gaps** — it prompted the boundary and validation unit tests that raised the suite from 60% to 78%.

Related Work

Prior work	Where we go further
Presidio, Macie, Google DLP — generic PII detection	No exclusion step before validation — structurally can't filter first
Schelter et al. — unit tests for data quality	We formalize the properties as metamorphic relations
Chen et al. — metamorphic testing for oracle-less systems	Applied here to PII detection over both detector types
Leucker & Schallhart — runtime verification	Our detector acts as a lightweight heuristic monitor
Gruschka et al. — GDPR challenges in big data	We provide a concrete, deployed, testable solution

Our contribution is a **testing methodology**, not a new detection algorithm — standard regexes and a phone validator made provably precise.

Limitations: Samples Are Not i.i.d.

Why

- Corpus positives = format/case variants of 26 positive seeds
- Corpus negatives instantiate 8 exclusion families from 52 negative seeds
- Confidence intervals bound performance **over these families**, not an arbitrary provider's data

What substantiates the bound

- 28 samples drawn directly from real production uploads
- 18 months in production with **zero** false-positive incidents reported

Claims hold within the families we tested — generalizing beyond food data is future work.

18 Months in Production, Zero False Positives

Contributions

- Exclusion filters applied **before** validation — closes an architectural gap generic tools structurally can't
- Four metamorphic relations generate the regression suite from 78 seeds
- Head-to-head Presidio comparison and mutation testing quantify the gains

Deployed in SOS Food



SOS Food (Horizon Europe, 2024–2027) extends FOODITY into closed-access settings. The detector has run continuously across both platforms for 18 months, serving **15 pilot organizations** — zero false-positive incidents reported to date.

Future work: ML-based NER for names and addresses, corpus expansion, formal evaluation of the keyword warning layer.



Thank you.

Questions welcome.

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Try it yourself: github.com/Modelio-R-D/SOSFood